Kingdom of Saudi Arabia
Ministry of Education
King Faisal University
College of Computer Sciences & Information Technology

Smart Door Lock System

by

Mohamed Abdullah Al mehaish (221433904)

Supervised by

Dr. Rama Samy

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1. Introduction

This project presents a Smart Door Lock System designed using Arduino Uno and embedded systems concepts. The system allows users to unlock the door by entering a password through a keypad. The entered password is verified by the Arduino, and the door opens automatically if the password is correct.

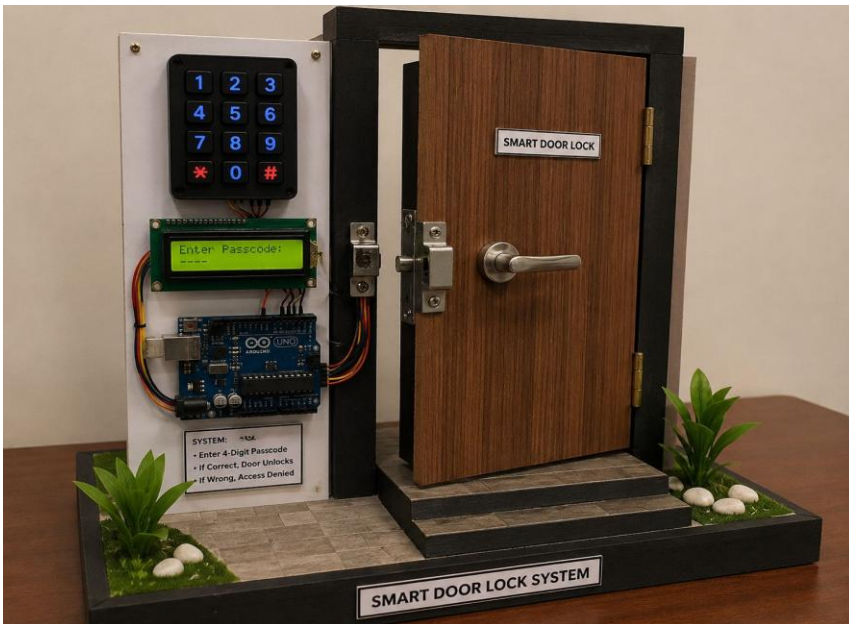

Figure 1: Complete Project Photo

2. Project Objectives

- ❖ Improve security using password authentication.
- ❖ Control the door electronically.
- ❖ Display system status using an LCD screen.
- ❖ Apply embedded systems concepts practically.

3. Components Used

- ❖ Arduino Uno
- ❖ LCD Display (16x2)
- ❖ 4x4 Keypad
- ❖ Servo Motor / Electronic Lock
- ❖ Jumper Wires
- ❖ Power Supply

4. System Working Principle

The system starts by displaying a password request message on the LCD screen. The user enters the password using the keypad. The Arduino compares the entered password with the saved password.

If the password is correct:

- The servo motor rotates.
- The door unlocks.
- A success message appears on the LCD screen.

If the password is incorrect:

- The LCD displays an error message.
- The door remains locked.

Table 1: States of the Project

No.	Condition	System Action	LCD Message
1	System Start	Arduino initializes all components	Enter Password
2	Correct Password	Servo motor unlocks the door	Access Granted
3	Door Open State	Door stays open for 5 seconds	Door Unlocked
4	Door Lock Again	Servo motor locks the door again	Door Locked
5	Incorrect Password	Door remains locked	Wrong Password

5. Wiring Diagram

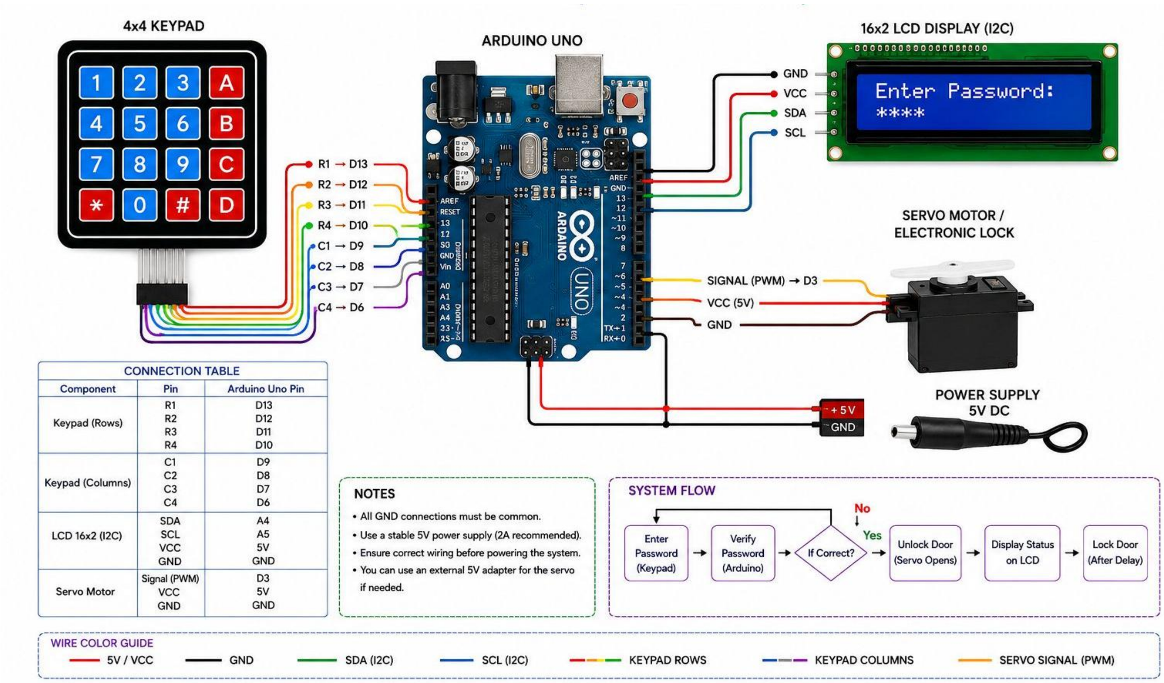

Figure 2: wiring Diagram using Arduino

6. Flowchart

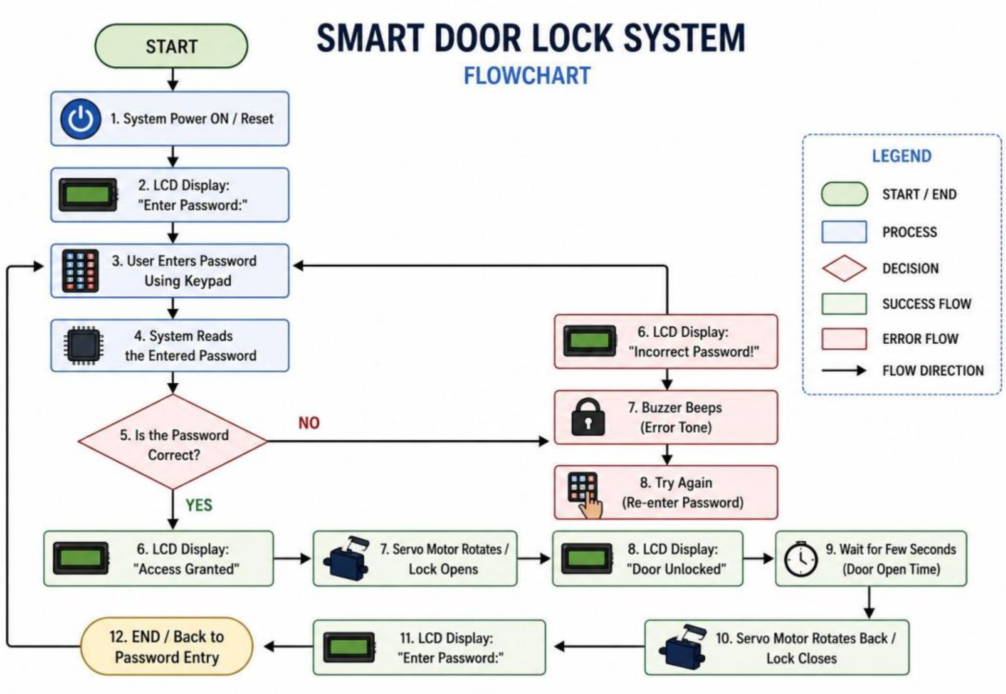

Figure 3: The Flowchart of the project

7. Source Code

```
#include <Keypad.h>

#include <LiquidCrystal.h>

#include <Servo.h>


// LCD Pins

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);


// Servo

Servo lockServo;


// Password

String password = "1234";

String inputPassword = "";


// Keypad Setup

const byte ROWS = 4;

const byte COLS = 4;


char keys[ROWS][COLS] = {

  {'1','2','3','A'},

  {'4','5','6','B'},

  {'7','8','9','C'},

  {'*','0','#','D'}
```

```
};

byte rowPins[ROWS] = {A0, A1, A2, A3};
byte colPins[COLS] = {A4, A5, 6, 7};

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

void setup() {

  lcd.begin(16, 2);

  lockServo.attach(9);

  lockServo.write(0); // Locked position

  lcd.setCursor(0,0);
  lcd.print("Smart Door Lock");

  delay(2000);

  lcd.clear();
  lcd.print("Enter Password");
}

void loop() {
```

```
char key = keypad.getKey();

if (key) {

  // Display *
  if (key != '#') {
    inputPassword += key;

    lcd.setCursor(inputPassword.length()-1,1);
    lcd.print("*");
  }

  // Check Password
  if (key == '#') {

    lcd.clear();

    if (inputPassword == password) {

      lcd.print("Access Granted");

      lockServo.write(90); // Unlock

      delay(5000);
```

```
      lockServo.write(0); // Lock again

      lcd.clear();
      lcd.print("Door Locked");

      delay(2000);

    } else {

      lcd.print("Wrong Password");

      delay(2000);
    }

    // Reset
    inputPassword = "";

    lcd.clear();
    lcd.print("Enter Password");
  }
 }
}
```

8. Future Improvements

- ❖ Add fingerprint authentication for higher security.
- ❖ Develop a mobile application to control the door remotely.
- ❖ Integrate Wi-Fi or Bluetooth connectivity.
- ❖ Add a buzzer alarm for incorrect password attempts.
- ❖ Use RFID card access for faster authentication.
- ❖ Store password data securely in memory.
- ❖ Add camera monitoring system for smart security.
- ❖ Improve the system using IoT technology.
- ❖ Add battery backup during power failure.
- ❖ Support multiple users with different passwords.

9. References

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" Available: https://www.arduino.cc/

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[3]Arduino Playground, "Keypad Library,
" Available: https://playground.arduino.cc/Code/Keypad/

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